

Maciej Czerniak

Data Science & AI Student

Machine Learning, Data Analytics & Applied AI

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🐙 GitHub

🌐 LinkedIn

📁 Portfolio

PROFILE

Data Science and Artificial Intelligence student at Breda University of Applied Sciences. Building applied AI and software systems across computer vision, geospatial analysis, robotics, LLM-backed query assistants, and Python package development. Currently working as a Student Assistant Developer on Tangible Landscape systems combining depth sensing, GRASS GIS, real-time sensor processing, and interactive visualization.

EXPERIENCE

•Breda University of Applied Sciences

Nov. 2024 – Present

Student Assistant Developer

Breda, Netherlands

- Developing Python-based Tangible Landscape systems connecting physical 3D models with computer vision, depth sensing, GIS tooling, and interactive visualization.
- Built workflows for real-time depth-camera processing, object detection, coordinate mapping, geospatial analysis, and visualization.
- Integrated and adapted sensor pipelines, including the transition from Xbox Kinect to Orbbec Femto Bolt.
- Implemented interactive geospatial features including tree-planting, viewshed analysis, and topography simulations.

EDUCATION

•Breda University of Applied Sciences

Sep. 2024 – Present

BSc Applied Data Science and Artificial Intelligence

Breda, Netherlands

- Relevant areas: machine learning, data analytics, statistics, computer vision, robotics, software engineering, databases, and applied AI projects.

•Zespół Szkół Budowlanych im. Eugeniusza Kwiatkowskiego

2012 – 2016

High School Diploma: Building Technician

Konin, Poland

PROJECTS

•Sign Language Recognition Package

Apr. 2026 - Jun. 2026

Python package refactor for model training, evaluation, testing, and deployment preparation

- Refactored notebook-based training code into a reusable Python package with configuration, data loading, preprocessing, training, evaluation, logging, and output persistence.
- Built the Azure ML training and deployment workflow, including training scripts, pipeline logic, model registration, endpoint testing, and deployment automation.
- Implemented MLflow tracking and Azure-based model lifecycle workflows to support reproducible training, evaluation, versioning, and controlled model promotion.
- Created GitHub Actions workflows for quality checks, Docker builds, Azure deployment, documentation deployment, and promoted-model rollout.

•Query Assistant

Jan. 2026 – Jan. 2026

LLM-backed assistant for converting natural language questions into SQL queries

- Built a prototype assistant that converts business questions into SQL queries using locally hosted LLMs.
- Connected the system to relational databases and tested generated queries against structured business data.
- Improved prompting and validation rules to reduce invalid SQL and make outputs more reliable.
- Used Python, SQLAlchemy, Pandas, MySQL/PostgreSQL/SQLite, Ollama, Streamlit, and prompt engineering.

•Root Inoculation Pipeline

Nov. 2025 – Jan. 2026

Computer vision and robotics pipeline for plant root analysis and robotic inoculation

- Developed multiclass image segmentation workflows for root, shoot, seed, and background classes.
- Built root system architecture extraction logic to identify plant instances, primary roots, and biologically relevant target points.
- Connected image-based analysis with simulated robotic control for an Opentrons OT-2 liquid-handling robot.
- Used Python, TensorFlow/Keras, OpenCV, image processing, evaluation metrics, PyBullet, and experiment tracking concepts.

TECHNICAL SKILLS

Languages: Python, SQL

AI & Data: machine learning, deep learning, computer vision, image segmentation, model evaluation, data preprocessing

Libraries: Pandas, NumPy, OpenCV, TensorFlow/Keras, PyTorch, scikit-learn, Matplotlib

MLOps & Tools: Azure ML, MLflow, Docker, GitHub Actions, CI/CD, Git, Poetry, pytest, FastAPI

Databases & GIS: PostgreSQL, MySQL, SQLite, GRASS GIS, Tangible Landscape